Physics-Based Simulink And Robot Bridge

Full-scale haul truck plant and Turtle robot handoff

Haulage Simulation Team

June 2026





What This Update Covers

Project direction

Move from a transparent speed-advisory simulation toward a model that can also show physical vehicle response and lab robot behavior.

- Keep the route and speed-policy logic modular.
- Add built-in MATLAB/Simulink physics blocks.
- Produce a concrete robot command output for lab testing.

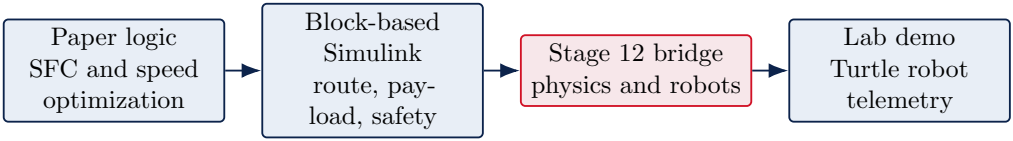
Main message

The ML model recommends a speed. The physical plant decides how the truck or robot actually responds.

This is a bridge model, not the final validated haul-truck digital twin.

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Where This Fits In The Framework



ML advisory

payload + grade class produces an efficient speed recommendation

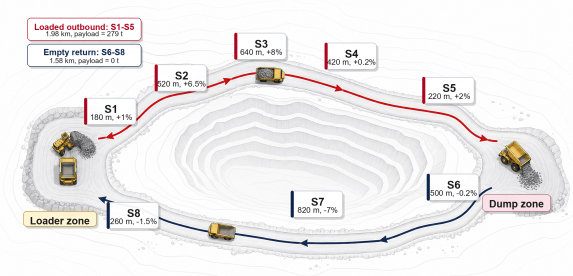
Full-scale plant

Simscape Driveline tests vehicle response to torque, grade, and resistance

Robot handoff

Robotics System Toolbox converts route ideas into /cmd_vel commands

Route Context



Open-pit demonstration route

- 8 route segments
- Loaded outbound: S1–S5
- Empty return: S6–S8
- Segment fields: distance, grade, speed limit, zone, payload

Why this matters

Grade, payload, and local limits change the physically reasonable speed command.

Route Values Used For The Open-Pit Demo

Seg.	Operating role	Start–End (m)	Length (m)	Grade (%)	Payload
1	Loader exit	0–180	180	1.0	279 t
2	Ramp up 1	180–700	520	6.5	279 t
3	Ramp up 2	700–1340	640	8.0	279 t
4	Bench connector	1340–1760	420	0.2	279 t
5	Dump approach	1760–1980	220	2.0	279 t
6	Return flat	1980–2480	500	-0.2	0 t
7	Ramp down	2480–3300	820	-7.0	0 t
8	Loader approach	3300–3560	260	-1.5	0 t

Editable route properties

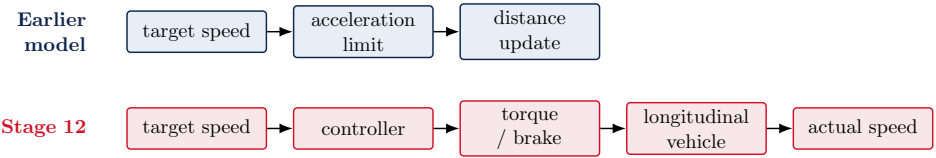
The block-based model stores route geometry, grade, payload, speed limit, zone code, and grade class as selectable scenario inputs.

Current scale

Total route distance is 3.56 km. The route is a demonstration scenario, not a surveyed mine road.

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Why Add A Built-In Longitudinal Vehicle Block?



What the older model does well

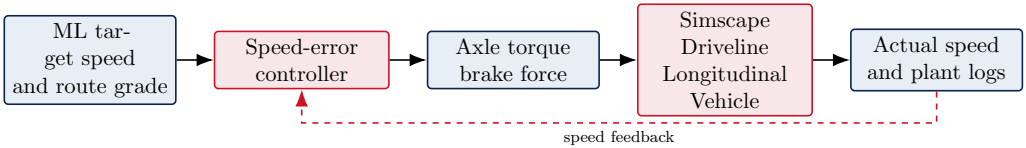
It is transparent and explains the speed optimization logic clearly.

What the physics block adds

It exposes mass, grade, rolling resistance, drag, torque, braking, and plant response.

Reasoning: a speed recommendation must pass through a physical vehicle before it becomes real motion.

Full-Scale Haul Truck Plant



Inputs

target speed, grade, wind, feedback speed

Plant

mass, rolling resistance, drag, tire radius

Outputs

speed, torque command, brake command, logs

Physics Represented In The Longitudinal Plant

Core balance

$$F_{\text{drive}} - F_{\text{brake}} - F_{\text{resist}} = ma$$

$$F_{\text{resist}} = F_{\text{grade}} + F_{\text{roll}} + F_{\text{drag}}$$

- Uphill grade increases required tractive effort.
- Rolling resistance scales with vehicle weight.
- Aerodynamic drag grows with speed squared.

Main approximations

$$F_{\text{grade}} \approx mg \cdot \text{grade}$$

$$F_{\text{roll}} \approx C_{\text{roll}} mg$$

$$F_{\text{drag}} \approx \frac{1}{2}\rho C_d A v^2$$

The built-in block manages the plant dynamics. Our model supplies scenario parameters and control commands.

Truck Parameters In The First Physics Bridge

Parameter	Value
Empty truck mass	150,000 kg
Loaded payload used	279,000 kg
Total loaded mass	429,000 kg
Tire rolling radius	1.8 m
Rolling coefficient	0.03
Drag coefficient	0.70
Frontal area	10 m^2

Current limitation

The first Simscape plant uses a fixed loaded mass and a stable loaded-outbound slice. It is not yet the full loaded-and-empty haul cycle.

Why this is still useful

It lets the team inspect how target speed, grade, torque, and vehicle speed interact before adding more detailed braking, retarding, engine, and tire models.

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Turtle Robot Kinematic Plant

Robotics System Toolbox block

The robot side uses the **Unicycle Kinematic Model**. It represents planar motion using forward velocity and heading rate.

$$\dot{x} = v\cos(\theta)$$

$$\dot{y} = v\sin(\theta)$$

$$\dot{\theta} = \omega$$

Why this maps to Turtle robots

Turtle-style differential-drive robots are commonly commanded with ROS velocity messages:

$$/cmd_vel$$

`geometry_msgs/Twist`

Model input	ROS field
v	`Twist.linear.x`
ω	`Twist.angular.z`

Robot Team Output

Main handoff file

`stage12_turtle_cmd_vel_schedule.csv`

This file is a time-indexed velocity schedule for robot playback.

CSV column	Use on robot
`time_s`	command timestamp
`linear_velocity_mps`	`Twist.linear.x`
`angular_velocity_radps`	`Twist.angular.z`
`segmentId`	route segment metadata
`grade_pct`	route grade metadata

How the team uses it

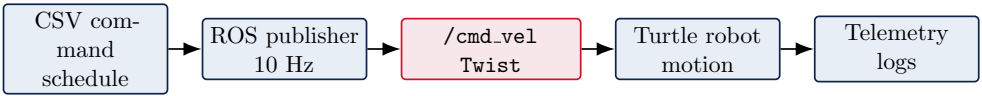
Publish each row to `/cmd_vel` at 10 Hz, then log the robot response.

Also included

- `stage12_turtle_route_waypoints.csv`
- `stage12_turtle_robot_handoff.json`
- `stage12_turtle_robot_handoff.md`

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How The Robot Test Should Be Run



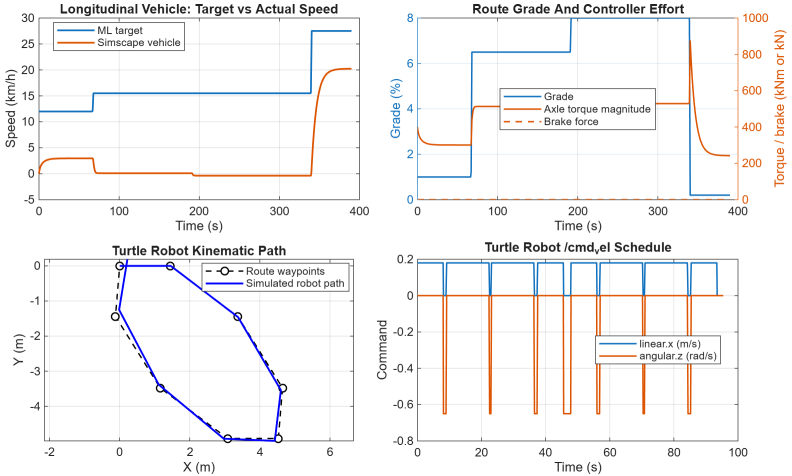
First safety run

Lift wheels or use blocks, test a short window, confirm direction, then move to open floor space with an emergency stop ready.

What to log

`/cmd_vel`, `/odom`, timestamps, battery voltage if available, and notes on floor surface/payload.

Current Stage 12 Dashboard



First longitudinal plant run, route grade/controller effort, Turtle path, and generated `/cmd_vel` schedule.

How To Reproduce The Current Model

MATLAB command

```
report = runStage12PhysicsKinematicsBridge();

open_system(report.modelPath)
```

Generated Simulink model

`simulink/HaulagePhysicsKinematicsBridge.slx`

Generated outputs

- `longitudinal_drive_cycle.csv`
- `simscape_signal_log.csv`
- `turtle_cmd_vel_schedule.csv`
- `turtle_route_waypoints.csv`
- `physics_kinematics_dashboard.png`

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What We Should Build Next

Full-scale haul truck side

- Replace time-indexed grade with distance-indexed road profile.
- Add loaded and empty plant variants.
- Add a controlled brake/retarder model.
- Add engine/drivetrain and tire detail where needed.

Robot side

- Run safe open-loop `/cmd_vel` playback.
- Log `/odom` and compare path tracking.
- Move from open-loop playback to Pure Pursuit.
- Add payload and surface-condition experiments.

Team alignment

The same modular idea is used at both scales: route segment, vehicle state, controller, safety check, plant response, and logged outputs.

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Clean Takeaway

1. Advisory layer
ML recommends a fuel-aware target speed from payload and grade context.

2. Physical layer
Simscape tests how a full-scale truck responds through torque, resistance, mass, and grade.

3. Robot layer
Robotics System Toolbox and ROS convert the same route idea into `/cmd_vel` commands.

A bridge from paper logic to engineering simulation to lab demonstration.

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Thanks for your attention
Questions?